250
0,00 0,50 1,00 1,50 2,00 2,50 3,00 3,50 4,00 4,50 5,00
% m/m sulphur in fuel
SO2(ppm)/CO2(%)
Distillate fuel
Residual fuel

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4 Correspondence between 65 (ppm 4/%) SO2/CO2 and 6.0 g/kWh is demonstrated by showing that their S/C ratios are similar. This requires the additional assumption of a brake-specified fuel consumption value of 200 g/kWh. This is an appropriate average for marine diesel engines. The calculation is as follows:

S/Cfuel = brake-specific SO2
(MWS / MWSO2)
BSFC
(% carbon in fuel / 100)

brake-specific SO2 = 6.0 g/kW-hr

MWS = 32.065 g/mol

MWSO2 = 64.064 g/mol

BSFC = 200 g/kW- hr

% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%

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SOURCE: MEPC.259(68).pdf
(MWS / MWSO2)
BSFC
(% carbon in fuel / 100)

brake-specific SO2 = 6.0 g/kW-hr

MWS = 32.065 g/mol

MWSO2 = 64.064 g/mol

BSFC = 200 g/kW- hr

% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%
(residual)

S/Cresidual fuel = 6.0
(32.065 / 64.064) = 0.01723 200
(87.17% / 100)

S/Cdistillate fuel = 6.0
(32.065 / 64.064) = 0.01765 200
(85.05% / 100)

Note 1: The S/C mass ratios calculated above, based on 6.0 g/kWh and 200 g/kWh BSFC, are both within 0.10% of the S/C mass ratios in the emissions table (Table 2). Therefore, 65 (ppm4/%) SO2/CO2 corresponds well to 6.0 g/kWh.

Note 2: The value of 6.0 g/kWh, hence the 200g/kWh brake-specified fuel consumption is taken from MARPOL Annex VI as adopted by the 1997 MARPOL Conference.

5 Thus, the working formulas are as follows:

For complete combustion = SO2 (ppm*) 65 CO2 (%*)

For complete combustion = SO2 (ppm*) 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)

* Note: gas concentrations must be sampled or converted to the same residual water content (e.g., fully wet, fully dry).

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SOURCE: MEPC.259(68).pdf
For complete combustion = SO2 (ppm*) 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)

* Note: gas concentrations must be sampled or converted to the same residual water content (e.g., fully wet, fully dry).

6 The following is the basis of using the (ppm 4/%) SO2/CO2 as the limit for determining compliance with regulation 14.1 or 14.4 of MARPOL Annex VI:

.1 This limit can be used to determine compliance from fuel oil burners that do not produce mechanical power.

.2 This limit can be used to determine compliance at any power output, including idle.

.3 This limit only requires two gas concentration measurements at one sampling location.

.4 There is no need to measure any engine parameters such as engine speed, engine torque, engine exhaust flow, or engine fuel flow.

.5 If both gas concentration measurements are made at the same residual water content in the sample (e.g., fully wet, fully dry), no dry- to-wet

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SOURCE: MEPC.259(68).pdf
speed, engine torque, engine exhaust flow, or engine fuel flow.

.5 If both gas concentration measurements are made at the same residual water content in the sample (e.g., fully wet, fully dry), no dry- to-wet conversion factors are required in the calculation.

.6 This limit completely decouples the thermal efficiency of the fuel oil combustion unit from the EGC unit.

.7 No fuel properties need to be known.

.8 Because only two measurements are made at a single location, transient engine or EGCS unit effects can be minimized by aligning signals from just these two analysers. (Note that the most appropriate points to align are the points where each analyser responds to a step change in emissions at the sample probe by 50% of the steady-state value.)

.9 This limit is independent of the amount of exhaust gas dilution. Dilution may occur due to evaporation of water in an EGC unit, and as part of an exhaust sampler's preconditioning system.

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APPENDIX 3

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APPENDIX 3

WASHWATER DATA COLLECTION

1 The washwater discharge criteria are intended to act as initial guidance for implementing EGC system designs. The criteria should be revised in the future as more data

becomes available on the contents of the discharge and its effects, taking into account any advice given by GESAMP.

2 Administrations should therefore provide for collection of relevant data. To this end, shipowners in conjunction with the EGC manufacturer are requested to sample and analyse samples of:

- inlet water (for background);
- water after the scrubber (but before any treatment system); and
- discharge water.

3 This sampling could be made during approval testing or shortly after commissioning and at about twelve-month intervals for a period of two years of operation (minimum of three samples). Sampling guidance and analysis should be undertaken by laboratories using EPA or ISO test procedures for the following parameters:

- pH

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SOURCE: MEPC.259(68).pdf
and at about twelve-month intervals for a period of two years of operation (minimum of three samples). Sampling guidance and analysis should be undertaken by laboratories using EPA